

Implementing Chatbot Architecture for a Feynman-Driven Educational Chatbot

Learn • Reason • Collaborate

HW0288 Assignment

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Shifting towards Education 4.0

Passive Learning



Relies primarily on listening, observing, and **passive information absorption**.



Active Learning



Engage students in the learning process
(Project Based Learning etc.) [1]



Active Learning: Core Benefits

1

Improves
Communication

3

Encourages
Critical Thinking [2]

5

Promotes
Collaboration

2

Enhances
Understanding

4

Boosts Academic
Performance
(STEM)



Active Learning: Implementation Challenges



Knowledge Gap

Lack of prior knowledge required for meaningful explanation.



Peer Evaluation

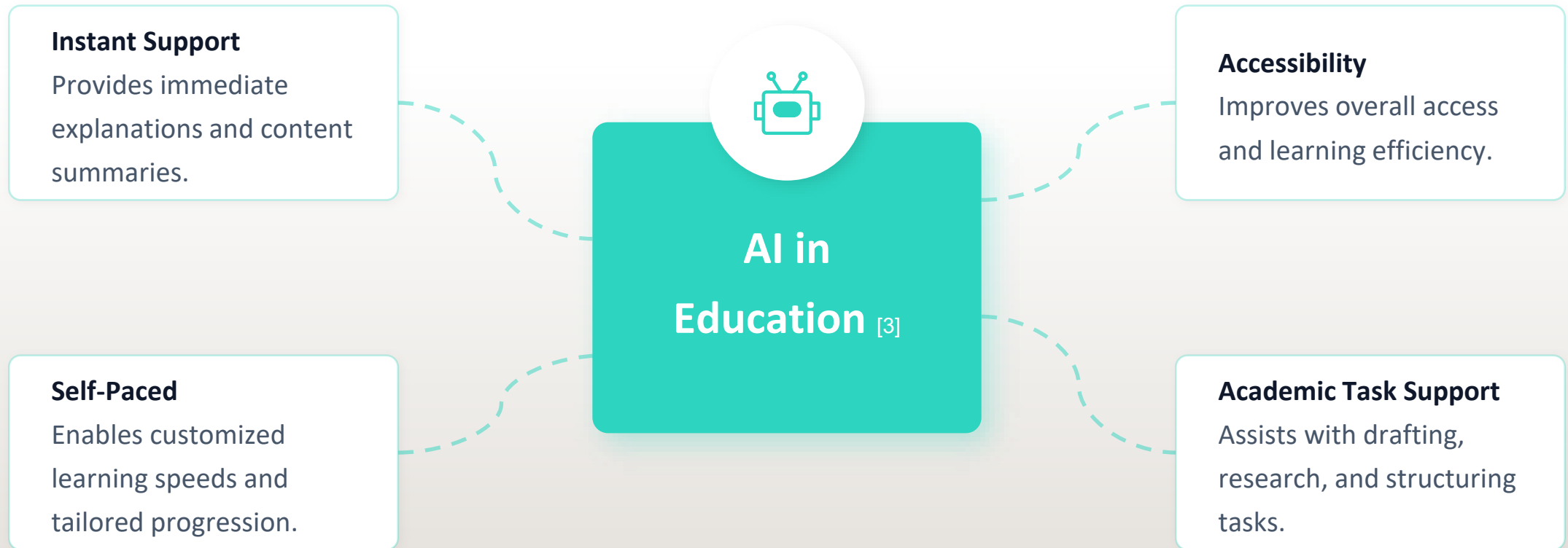
Ineffective or inaccurate feedback during peer-to-peer assessment.



Resource Heavy

Requires significant time investment and active instructor supervision. [2]

AI in Education: Key Benefits



AI in Education: Critical Limitations



Passive Learning

Over-reliance encourages students to passively absorb AI-generated answers rather than thinking critically.



Hallucinations

Generative models can confidently produce inaccurate, biased, or entirely fabricated information.



Oversimplification

Complex, nuanced topics are often reduced to generic summaries, stripping away necessary depth.



Lack of Validation

AI often fails to challenge the user's misconceptions or validate the depth of their actual understanding.

The Core Problem

Active Learning

✓ Highly Effective

Drives deep cognitive engagement and conceptual mastery.

✗ Hard to Scale

Requires intense instructor supervision and time.

VS

AI Assistants

✓ Highly Scalable

Provides instant, on-demand support for any number of students.

✗ Passive Learning

Encourages relying on generated answers rather than reasoning.

Solution: Feynman Technique

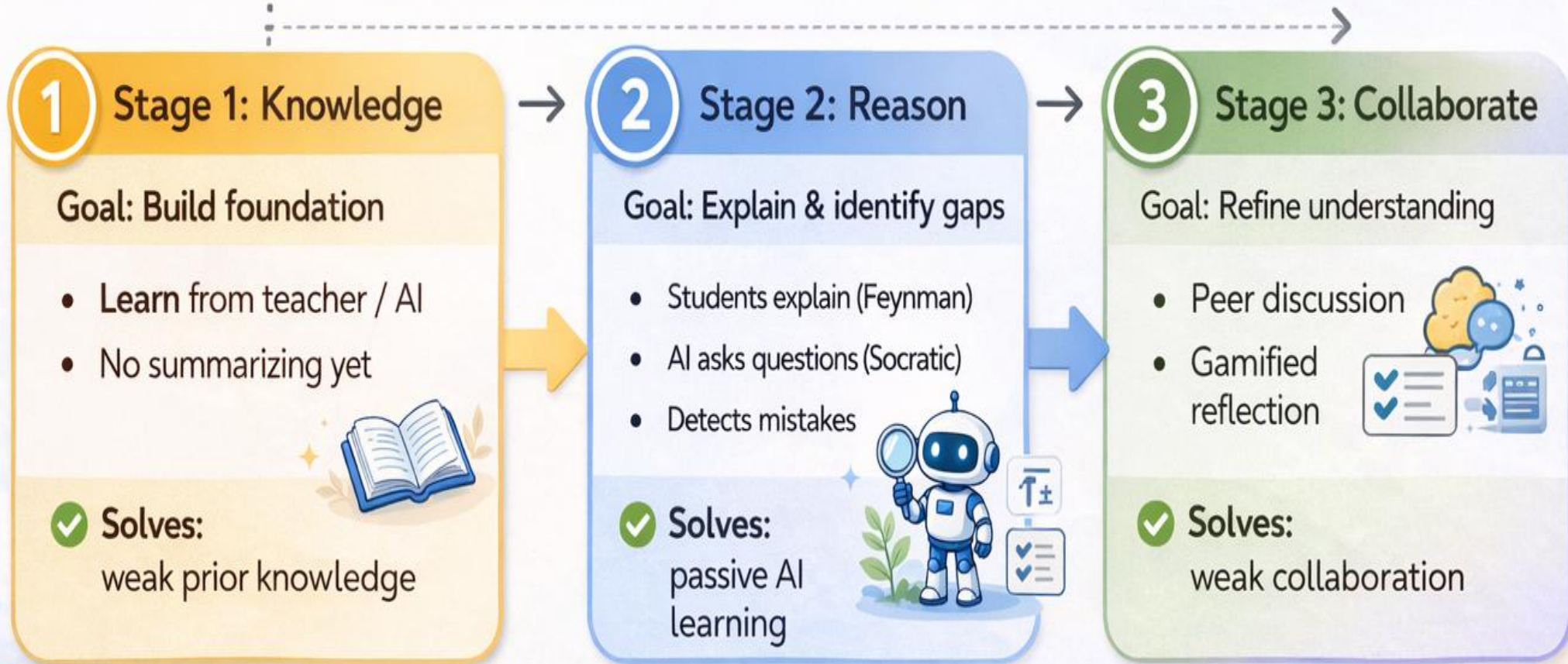
1 Pick a concept

2 Explain Simply

3 Identify gaps

4 Simplify & refine

Feynman Critical Thinking Architecture



References

- [1] M. Adeoye, "View of from Struggle to Success: The Feynman Techniques' Revolutionary Impact on Slow Learners," Undiksha.ac.id, 25 Oct. 2023. [Online]. Available: <https://ejournal.undiksha.ac.id/index.php/TSCJ/article/view/69681/29104>. [Accessed: 25-Feb-2026].
- [2] A. A. Rahman et al., "Literature Review on the Benefits and Challenges of Active Learning on Students' Achievement," Cypriot Journal of Educational Sciences, vol. 17, no. 12, 31 Dec. 2022. [Online]. Available: <https://doi.org/10.18844/cjes.v17i12.8133> . [Accessed: 24-Feb-2026].
- [3] A. Upadhyay et al., "Influence of LLMs on Learning and Teaching in Higher Education," Social Science Research Network, 1 Jan. 2024. [Online]. Available: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4716855, <https://doi.org/10.2139/ssrn.4716855>. [Accessed: 21-Feb-2026].